What Can Each Study Claim?

AP Statistics · Unit 1 · Topic 1.10 · B: 2026-09-28 · E: 2026-10-05 · TEACHER KEY

CED TETHER

- **Skill 1.C** — Describe an appropriate method for gathering and representing data.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **EU DAT-2** — The way we collect data influences what we can and cannot say about a population.
- **LO DAT-2.A** — Identify the type of a study. [Skill 1.C]
- **EK DAT-2.A.1** — A population consists of all items or subjects of interest.
- **EK DAT-2.A.2** — A sample selected for study is a subset of the population.
- **EK DAT-2.A.3** — In an observational study, treatments are not imposed. Investigators examine data for a sample of individuals (retrospective) or follow a sample of individuals into the future collecting data (prospective) in order to investigate a topic of interest about the population. A sample survey is a type of observational study that collects data from a sample in an attempt to learn about the population from which the sample was taken.
- **EK DAT-2.A.4** — In an experiment, different conditions (treatments) are assigned to experimental units (participants or subjects).
- **LO DAT-2.B** — Identify appropriate generalizations and determinations based on observational studies. [Skill 4.A]
- **EK DAT-2.B.1** — It is only appropriate to make generalizations about a population based on samples that are randomly selected or otherwise representative of that population.
- **EK DAT-2.B.2** — A sample is only generalizable to the population from which the sample was selected.
- **EK DAT-2.B.3** — It is not possible to determine causal relationships between variables using data collected in an observational study.

Essential question: How do selection and assignment determine the conclusions a study can support?

DOK-3 skill: 4.A · **FRQ pattern:** `observational-vs-experiment-what-each-can-claim`

DOK rationale: Part (c) weighs two study plans against a causal claim, connects self-selection to a concrete lurking variable, and separately bounds what random assignment among volunteers can establish and generalize.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask for a plan choice before defining study types.
- Begin the video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and separate assignment from selection.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose Plan A or Plan B and name one design feature in a first take.
- Complete the video follow-along about populations, samples, observational studies, and experiments.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Who decides whether a student receives tutoring in each plan?
- How could motivation affect both the group a student joins and the score later measured?
- What does the coin flip balance, and among whom?
- Were the volunteers selected by chance from all students? What conclusion does that limit?

Adult role

Solo teacher. Circulate 5 pairs. Ask students to label each arrow as selection or assignment before accepting a claim about cause or scope.

[WARN] WATCH FOR

- Calling any comparison of two groups an experiment.
- Naming a lurking variable without linking it to both explanatory and response variables.
- Assuming random assignment also makes volunteers representative of all students.

SCORING PART (c) — E / P / I

Expected elements

- **selects-randomized-plan** (required) — Selects Plan B and identifies random assignment, not volunteering, as the feature supporting causation
- **explains-plan-a-alternative** (required) — Explains how a named self-selection or lurking variable could produce Plan A's difference
- **bounds-causal-scope** (required) — Limits the causal conclusion to the volunteers studied unless broader representativeness is justified
- **separates-assignment-from-selection** (required) — Distinguishes random assignment for cause from random selection or representativeness for generalization

E	Chooses Plan B because of random assignment, gives a linked lurking-variable explanation for Plan A, and correctly limits generalization beyond the volunteers.
P	Chooses Plan B and mentions random assignment but gives an unlinked lurking variable or omits the generalization limit, or states the two limits without clearly separating selection from assignment.
I	Claims Plan A establishes cause, says volunteering makes Plan B representative, or gives no design-based reason.

Common mistakes

- Calling Plan A an experiment because it compares two groups
- Treating the coin flip as random selection from the school instead of random assignment among volunteers
- Naming motivation without explaining how it relates to both attendance and scores
- Generalizing Plan B to all students solely because it is an experiment

[STAR] TODAY'S PROBLEM — annotated

Suppose a student wants to know whether a new tutoring center raises test scores.

Plan A: Compare this year's test scores for students who choose to attend the center with scores for students who do not attend.

Plan B: Recruit 40 student volunteers, then use a coin flip to assign each volunteer either to attend the center or not to attend. Compare their later test scores.

Two plans for studying tutoring

Plan	Who?	Groups?	Response
A	Students	Choose	Scores
B	40 volunteers	Coin flip	Scores

FIRST TAKE (5 min)

Which plan gives stronger evidence that the center raises scores? Name the feature behind your choice.
Key: Not graded. Listen for students who say Plan B is stronger merely because it is random without naming what is randomized.

Rules callout "SELECTION, ASSIGNMENT, AND SCOPE (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Identify Plan A as an observational study or experiment, and do the same for Plan B. State why for each.

Key: Plan A is observational because students choose whether to attend; no treatment is assigned. Plan B is an experiment because attendance is assigned to volunteers by a coin flip.

DOK 2 **(b)** Name one lurking variable that could make Plan A's groups differ, and explain how it could be related to both attendance and test score.

Key: One example is motivation: more motivated students may be more likely to choose tutoring and may also study more or perform better even without the center. Prior achievement, available time, or family support are acceptable if both links are explained.

DOK 3 **(c)** Which plan could support the claim that the center raises scores? Cite the design feature that settles it and explain what Plan A's result could mean instead. Then state one conclusion Plan B still could not support about students beyond the volunteers.

Key: Plan B could support a causal claim for these volunteers because random assignment tends to balance lurking variables between the attendance and no-attendance groups. Under Plan A, a score difference could be due to motivation or another preexisting difference rather than tutoring. Because the 40 participants volunteered rather than being randomly selected from a named population, Plan B cannot automatically generalize the effect to all students at the school or to students elsewhere.

EXIT

One thing the video changed about my first take: ____________________